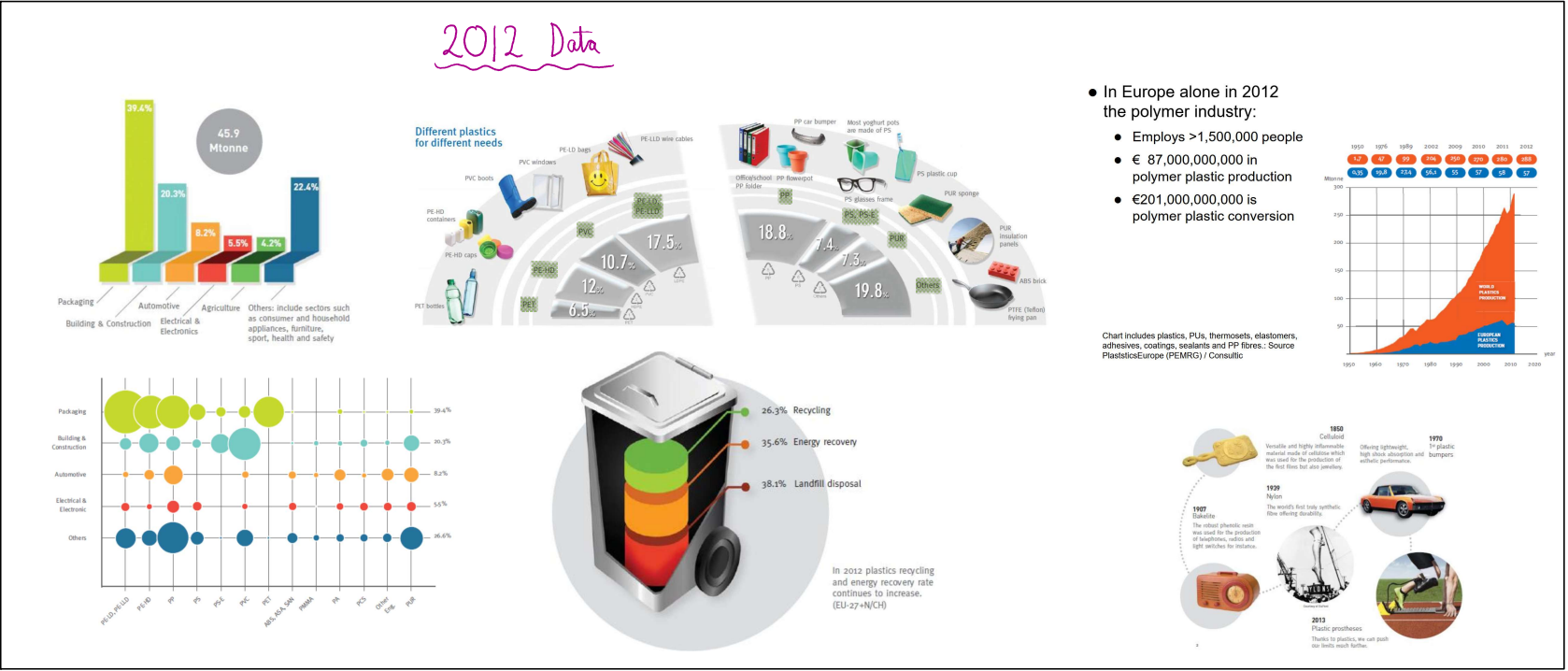


Polymers

Characteristics

- Low moduli (~1000x less than metals)
- ~5x lighter than metals.
- Some nearly as strong as metals
- Large elastic deflections
- Creep easily, even at room temp, so over time may reach permanent set if under load.
- Properties depend on temperature
- NONE have useful strength above 200°C
- Easy to shape / form
- Large elastic deflections allow for snap fixtures
 - Fast & Cheap assembly
- No finishing operations needed if mould is accurate
- Corrosion resistance
- Low coefficients of friction



Thermoplastics

- Predominantly linear molecules, with degree of branching and side chains.
- Can be melted and caused to flow repeatedly.
- Comparatively poor thermal properties
- Typically supplied as pellets; converted via thermal process to form shape needed.

E.g.

- Nylons
- Polyethylene (PE)
- ABS
- PET
- Polypropylene (PP)

Thermosetting Polymers

- Form a 3-D network which cannot be melted
- Often in form of liquid or solution that can be moulded; transforms from a liquid (typically a partially reacted polymer) to a macromolecule in a mould.
- Involves cross-linking ⇒ E.g. Unsaturated polyesters usually react with styrene (solvent) forming Copolymer of polyester; Polystyrene
- Relatively hard for polymers; Good temp. resistance.

E.g.

- Epoxy resins
- Unsaturated polyester

Processing

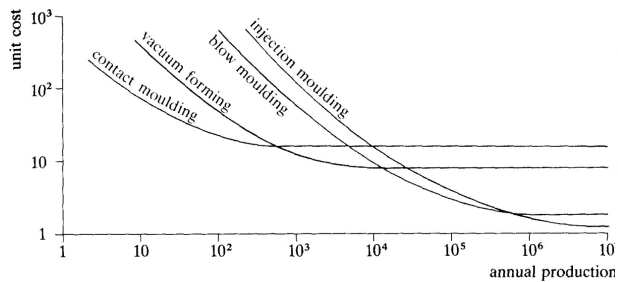
- Usually a moulding operation involving shape changes in liquid state
- At much lower temps. than those required for metals.
- Much lower pressures, hence lower tooling costs compared to metals

Thermosets

- Solidifies by virtue of chemical reaction where liquid transforms to solid by process called curing - Often requires heat or hot mould.
- At processing temps., viscosity is often very low. (like water)

Thermoplastics

- Solidifies by cooling in a mould
- Generally require cold / cool moulds
- At processing temps., viscosity is much higher. (like treacle)



- As these costs have to be recovered from the production of parts, the cost burden per part for injection moulding is very high if the production volume is low.
- However, the cost of producing each part if the capital costs are ignored is much less for injection moulding than other processes due to a rapid production and process automation.

Injection Moulding

- This process is designed to facilitate high speed processing of plastics into parts (which may be intricate).
- A feed unit introduces the polymer into the barrel.
- This barrel contains a screw that transports the granules into a chamber.
- The screw does work on the polymer which heats it up and induces melting - a process assisted by additional heating as necessary.

Extrusion Processes

- This is similar to the metal forming process and involves forcing a molten polymer through a die, chilling rapidly on exit to preserve the cross section and allowing the process to continue indefinitely.

Blow Moulding

- Blow moulding was originally developed for glasses and adapted successfully to polymers.
- The material (the parison) is supplied hot in tubular form enclosed by the split mould, and forced to adopt the shape by air (or gas) pressure. The process is ideal for producing simple shapes, like containers where the surface finish is only critical on the outside.
- The process is cheap as tooling costs are low as the pressures are not high and only small clamping forces required.

Vacuum Forming

- Here the required shape is formed using air pressure. A polymer sheet is drawn as a hot formable sheet onto a tool via the application of a vacuum.
- Again the process is cheap with a low cost tool due to the low pressures. The material requirements for the polymer will be similar to that required for blow moulding.
- The complexity of shape that can be made in this way is obviously limited but large shapes can be made economically.

Blown Film

- An extruder is fitted with an annular die, pointing upwards.
- This tube is blown up with air, and is drawn up in continuous process. The air is contained in the bubble by collapsing rolls at the top.
- Like blow moulding this induces a bi-axial orientation, which is then made permanent by the crystallisation process which fixes the orientation in place.
- This process can form thinner sheets of bags. Bags need to have the edges cut off and require rewinding into two different rolls. The bags require heat sealing at regular intervals and regular perforations to permit tear off or cutting.

Rotational Moulding

- Some simple shapes may be produced by rotating a large heated mould with a charge of polymer.
- The polymer inside the mould melts and coats the inside of the mould.
- This will then be allowed to cool whilst still rotating allowing a uniform build up of polymer and ultimately a part which has conformed to the mould tool.

- ### Other Forms of Injection Moulding
- Gas assisted injection moulding, which is used to produce hollow parts. As the gas is injected into the mould cavity.
 - Co-injection moulding, is used to mould multiple polymers into a single part.
 - Slush is a moulding process where multiple injection barrels are used with a single tool. Once the injection cycle is completed, the pressure is isolated before melt freezing occurs. This introduces a preferred and controlled orientation, reducing voids but is expensive and can damage the mould.

Gas Injection - External

The shut off valve closes. A micro-thin layer of gas is injected between one surface of the plastic and the adjacent mould cavity surface. Using the gas to apply pressure to the plastic, which is still molten. The gas progresses the molten plastic into the extremities of the cavity.

Gas Injection - Internal

The shut off valve closes. A micro-thin layer of gas is injected between one surface of the plastic and the adjacent mould cavity surface. Using the gas to apply pressure to the plastic, which is still molten. The gas progresses the molten plastic into the extremities of the cavity.

Calcium carbonate filler

CaCO₃ has been shown to improve certain mechanical and thermal properties of the resulting polymer composite such as increased thermal resistance [29], impact resistance [39], and elastic resistance [40]. CaCO₃ filled polypropylene in particular has been researched for its effect on yield strength [41], young's modulus [42], volume strain [43], tensile strength [44,45], impact toughness [46], flexural properties [46], and rheological behaviour [47,48]. It has been found that as filler percentage of CaCO₃ within the formulation increases, the potential for issues during processing as well as reduced flexural toughness and strength of the resulting composites could occur [35].

From
<<https://www.sciencedirect.com/science/article/pii/S2590048X24000360>>

Table 1. Yield strength and Young's modulus of polypropylene composites filled with calcium carbonate (CaCO₃) ranging from 0% to 40% at 5% increments.

Percentage CaCO ₃ filler	Yield strength (MPa)	Yield strength standard deviation (MPa)	Young's modulus (MPa)	Young's modulus standard deviation (MPa)
0%	17.68	1.2	733	72.3
5%	17.26	0.5	818	23.4
10%	17.60	0.7	925	47.5
15%	15.86	0.1	942	55.5
20%	15.17	1.0	961	80.8
25%	16.14	0.4	1009	29.2
30%	16.02	1.2	1168	78.9
35%	15.46	1.0	1205	75.2
40%	12.73	0.2	1240	28.9